

UChicago REU Notes Week 2

Zih-Yu Hsieh

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1 Review: Symmetric Monoidal Categories

Definition 1.1. A category $\mathcal{V}$ is a *monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (- \otimes -) \otimes - \xrightarrow{\sim} - \otimes (- \otimes -)$ of functors $\mathcal{V} \times \mathcal{V} \times \mathcal{V} \xrightarrow{\sim} \mathcal{V}$. (So, $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ in a canonical way).
- The *left unit* $\lambda : \kappa \otimes - \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ ($1 \otimes X \cong X$ in a canonical way).
- The *right unit* $\rho : - \otimes \kappa \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ ($X \otimes 1 \cong X$ in a canonical way).

They also satisfy two extra axioms:

- *Triangle Axiom:*

$$\begin{array}{ccc} (X \otimes 1) \otimes Y & \xrightarrow{a_{X,1,Y}} & X \otimes (1 \otimes Y) \\ \searrow \rho_X \otimes \text{id}_Y & & \swarrow \text{id}_X \otimes \lambda_Y \\ & X \otimes Y & \end{array}$$

- *Pentagon Axiom:*

$$\begin{array}{ccc} ((X \otimes Y) \otimes Z) \otimes W & \xrightarrow{a_{X \otimes Y, Z, W}} & (X \otimes Y) \otimes (Z \otimes W) \\ \downarrow a_{X,Y,Z} \otimes \text{id}_W & & \downarrow a_{X,Y,Z \otimes W} \\ (X \otimes (Y \otimes Z)) \otimes W & & X \otimes (Y \otimes (Z \otimes W)) \\ \searrow a_{X,Y \otimes Z, W} & & \swarrow \text{id}_X \otimes a_{Y,Z,W} \\ & X \otimes ((Y \otimes Z) \otimes W) & \end{array}$$

Based on *MacLane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \rightarrow P_2$, where P_i are (possibly) different orders of performing product on $X_1, \dots, X_n \in \mathcal{V}$. For reference, cf. *Category Theory for Working Mathematicians*.

This guarantees all monoidal category to be equivalent to a *strict monoidal category*, namely all a, λ, ρ above are “identity transformations”, or $1 \otimes X = X, X \otimes 1 = X$, and $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, so WLOG one can assume the product $\otimes$ is associative.

Definition 1.2. A monoidal category $\mathcal{V}$ is *braided*, if there is a natural isomorphism $B : {}_{-1} \otimes {}_{-2} \xrightarrow{\sim} {}_{-2} \otimes {}_{-1}$ (so $B_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ is an isomorphism), together with the *Hexagon axioms* satisfied:

$$\begin{array}{ccc}
(X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) \xrightarrow{B_{X,Y \otimes Z}} (Y \otimes Z) \otimes X \\
\downarrow B_{X,Y} \otimes \text{Id}_Z & & \downarrow a_{Y,Z,X} \\
(Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) \xrightarrow{\text{Id}_Y \otimes B_{X,Z}} Y \otimes (Z \otimes X) \\
\\
X \otimes (Y \otimes Z) & \xleftarrow{a_{X,Y,Z}} & (X \otimes Y) \otimes Z \xrightarrow{B_{X \otimes Y,Z}} Z \otimes (X \otimes Y) \\
\downarrow \text{Id}_X \otimes B_{Y,Z} & & \downarrow a_{Z,X,Y} \\
X \otimes (Z \otimes Y) & \xleftarrow{a_{X,Z,Y}} & (X \otimes Z) \otimes Y \xrightarrow{B_{X,Z} \otimes \text{Id}_Y} (Z \otimes X) \otimes Y
\end{array}$$

$\mathcal{V}$ is a *symmetric monoidal category*, if $B_{Y,X} = B_{X,Y}^{-1}$ for all $X, Y \in \mathcal{V}$.

The two hexagons are uniqueness of 3-cycle permutations on three consecutive objects in a product, hence together with B serving as transpositions, it guarantees unique isomorphism between all permutations of finite products, i.e., it imposes that there is a single way of doing such permutation (so there's no “nontrivial braiding”, when $B_{Y,X} \circ B_{X,Y} : X \otimes Y \rightarrow Y \otimes X \rightarrow X \otimes Y$ is not identity). From my naive understanding of coherence theorem, WLOG we can again treat $X \otimes Y = Y \otimes X$.

Example 1.3. Everyone loves unital commutative ring R , so I use category of R -modules as example :)

Equipped with tensor product $\otimes_R$, and R itself as tensor unit (up to isomorphism), we see $R\text{-Mod}$ forms a symmetric monoidal category due to following natural isomorphisms:

$$\left\{ \begin{array}{ll} (M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L) & , \quad (m \otimes n) \otimes \ell \mapsto m \otimes (n \otimes \ell) \\ R \otimes_R M \cong M & , \quad r \otimes m \mapsto rm \\ M \otimes_R R \cong M & , \quad m \otimes r \mapsto rm \\ M \otimes_R N \cong N \otimes_R M & , \quad m \otimes n \mapsto n \otimes m \end{array} \right.$$

Now, observe that $R\text{-Mod}$ is self-enriched, meaning $\text{Hom}_R(-, -) : R\text{-Mod}^{\text{op}} \times R\text{-Mod} \rightarrow R\text{-Mod}$ is a bifunctor (also called an *internal hom functor*). Moreover, there's the tensor-hom adjunction:

$$\text{Hom}_R(M \otimes_R N, L) \cong \text{Hom}_R(M, \text{Hom}_R(N, L)), \quad f \mapsto (m \mapsto f(m \otimes -))$$

This indicates $(- \otimes_R N \dashv \text{Hom}_R(N, -))$ is an adjoint pair, guaranteeing $- \otimes_R N$ commutes with all colimits. For instance, $(M \oplus L) \otimes_R N = (M \otimes_R N) \oplus (L \otimes_R N)$. This gives rise to the following definition:

Definition 1.4. $\mathcal{V}$ is a *symmetric monoidally closed category*, if there exists an *internal hom functor* $\mathcal{V}(-, -) : \mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$, such that the adjunction $(- \otimes X \dashv \mathcal{V}(X, -))$ holds for all $X \in \mathcal{V}$.

$\mathcal{V}$ is a *symmetric monoidally cocomplete category*, if it's cocomplete, together with $- \otimes X, X \otimes -$ preserves colimits for all $X \in \mathcal{V}$.

If people knows about Yoneda Lemma, welcome to look into the last section, which strengthen the isomorphism in monoidal closeness to natural isomorphism $\mathcal{V}(- \otimes X, -) \cong \mathcal{V}(-, \mathcal{V}(X, -))$ on the bifunctors $\mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$.

Example 1.5. The category of unital commutative rings $\mathbf{CRing}$ with $\otimes_{\mathbb{Z}}$ as monoidal product is in fact not monoidally closed. Suppose the contrary there exists such bifunctor $C : \mathbf{CRing}^{\text{op}} \times \mathbf{CRing} \rightarrow \mathbf{CRing}$, such that the adjunction holds, then the following isomorphism holds for any rings $R, S \in \mathbf{CRing}$:

$$\text{Hom}_{\mathbf{CRing}}(R, S) \cong \text{Hom}_{\mathbf{CRing}}(\mathbb{Z} \otimes_{\mathbb{Z}} R, S) \cong \text{Hom}_{\mathbf{CRing}}(\mathbb{Z}, C(R, S)) = \{*\}$$

The final implication follows from the fact $\mathbb{Z} \in \mathbf{CRing}$ is initial. Yet, here's a counterexample: $\text{Hom}_{\mathbf{CRing}}(\mathbb{C}, \mathbb{C})$ as at least two maps, including identity ($z \mapsto z$) and conjugation ($z \mapsto \bar{z}$), so we reach a contradiction.

Notice this is distinct from $\mathbf{CRing}$ being monoidally cocomplete: Coequalizer exists in $\mathbf{CRing}$ (by quotienting out the ideal generated by difference of outputs), and $\otimes_{\mathbb{Z}}$ is the coproduct in $\mathbf{CRing}$, hence all colimits exist (cf. *Limit Existence Theorem*). Moreover, colimits commutes with all colimit, in particular so does $\otimes_{\mathbb{Z}}$.

Remark 1.6. From now on, we assume either $\mathcal{V}$ is monoidally cocomplete. If suitable, also assume internal hom functor exists.

One reason to assume this, is to make sure some of the construction makes sense: In the literature professor May (implicitly) assumes monoidal product commutes with colimits, and occasionally use internal hom as an example of operad.

Notation 1.7. Let Σ_j denotes the j th symmetric group for all $j \in \mathbb{N}$. In particular, $\Sigma_0 = \Sigma_1 = \{1\}$ as trivial group.

Let Σ denote the category of finite numbers, with objects being $\mathbf{j} := \{1, \dots, j\}$, and homset

$$\text{Hom}_{\Sigma}(\mathbf{i}, \mathbf{j}) = \begin{cases} \emptyset & , \quad i \neq j \\ \Sigma_j & , \quad i = j \end{cases}$$

Also, let $\emptyset \in \mathcal{V}$ be the initial object. Since initial object is a colimit, $\emptyset$ exists, moreover $\emptyset \otimes X \cong \emptyset$ for all $X \in \mathcal{V}$ once assume monoidal cocompleteness.

2 Intro: Operads

As an example, we'll start with category of compactly generated based space $\mathcal{T}^*$, with cartesian product $\times$ as monoidal product, singleton $\{*\}$ as monoidal unit.

Example 2.1. Given $X \in \mathcal{T}^*$, define $\mathcal{E}_X(j) := \text{Hom}(X^j, X)$ for each $j \in \mathbb{N}$.

Let natural numbers $j = j_1 + \dots + j_k$, each $g_i \in \mathcal{E}_X(j_i)$ defines a map $g_i \circ \pi_i : X^j = \prod_{i=1}^k X^{j_i} \rightarrow X$, hence $g_1, \dots, g_k$ induces a product map $\Pi g_i : X^j \rightarrow X^k$. Now, given any $f \in \mathcal{E}_X(k)$, $f \circ g : X^j \rightarrow X^k \rightarrow X$ is an element in $\mathcal{E}_X(j)$. This induces the following structure map:

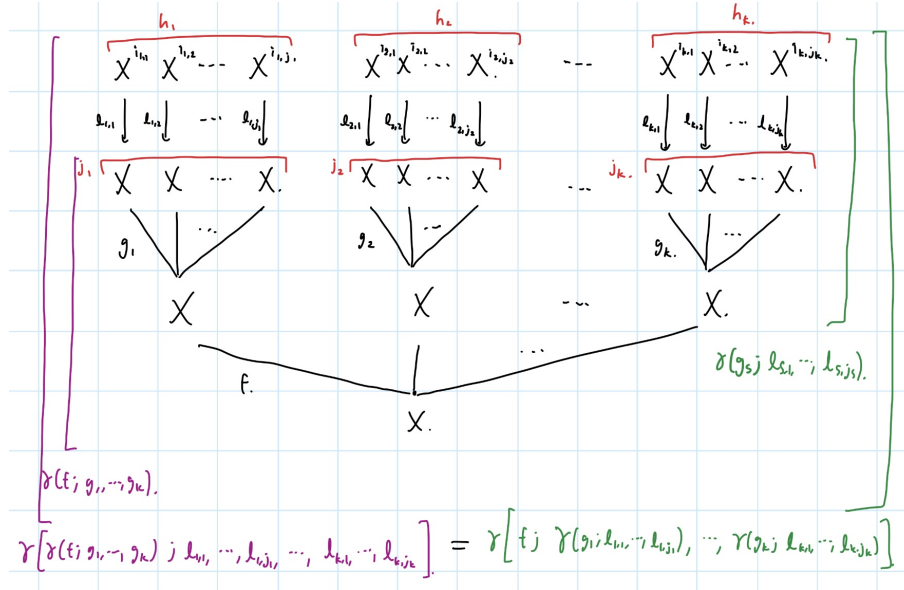
$$\gamma : \mathcal{E}_X(k) \times \prod_{i=1}^k \mathcal{E}_X(j_i) \rightarrow \mathcal{E}_X(j), \quad \gamma(f; g_1, \dots, g_k) := f \circ \Pi g_i$$

There is also a canonical choice of *unit*, namely $\eta : \{*\} \rightarrow \mathcal{E}_X(1)$ by $* \mapsto \text{id}_X$.

Unit here represents an *identity operation*, as one can verify the following two diagram commutes:

$$\begin{array}{ccc} \mathcal{E}_X(j) \times \{*\}^j & \xrightarrow{\sim} & \mathcal{E}_X(j) \\ \downarrow \text{id}_{\mathcal{E}_X(j)} \times \eta^j & \nearrow \gamma & \\ \mathcal{E}_X(j) \times \mathcal{E}_X(1)^j & & \end{array} \quad \begin{array}{ccc} \{*\} \times \mathcal{E}_X(j) & \xrightarrow{\sim} & \mathcal{E}_X(j) \\ \downarrow \eta \times \text{id}_{\mathcal{E}_X(j)} & \nearrow \gamma & \\ \mathcal{E}_X(1) \times \mathcal{E}_X(j) & & \end{array}$$

It also satisfies a giant associativity law, for demonstration purpose I'll draw a tree diagram instead. Given $k, j = j_1 + \dots + j_k$, each j_s associates with $h_s = i_{s,1} + \dots + i_{s,j_s}$, and $i := h_1 + \dots + h_k$. Then, the following combination diagram has no ambiguity, given $f \in \mathcal{E}_X(k)$, $g_s \in \mathcal{E}_X(j_s)$, and $\ell_{s,t} \in \mathcal{E}_X(i_{s,t})$:



Finally, there is a natural right Σ_j -action on $\mathcal{E}_X(j)$, by pre-permuting the input entries. Which, permutation in $\mathcal{E}_X(k)$ is equivalent to “permuting orders” of blocks $(j_1, \dots, j_k) = \{\{1, 2, \dots, j_1\}, \dots, \{j_1 + \dots + j_{k-1} + 1, \dots, j_1 + \dots + j_{k-1} + j_k\}\}$ in $\{1, \dots, j\}$, and each Σ_{j_s} -action on $\mathcal{E}_X(j_s)$ is equivalent to a permutation only on each block of $(j_1, \dots, j_k)$.

Intuitively, each $\mathcal{E}_X(j)$ encodes all possible way of operating on j points of X (a j -ary operation on X), while each $\mathcal{E}_X(k)$ also encodes operation of combining k (possibly distinct) operations into another operation, in an associative way.

This is called the *Endomorphism Operad of X* . In general, if $\mathcal{V}$ is monoidally closed, the internal hom functor provides a construction of such endomorphism operad in $\mathcal{V}$.

Definition 2.2. Given symmetric monoidal category $\mathcal{V}$ (use κ to denote unit), an Σ -operad $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j \in \mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j -action (a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \rightarrow \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) \rightarrow \mathcal{C}(j_1 + \dots + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{array}{ccc} \mathcal{C}(j) \otimes \kappa^{\otimes j} & \xrightarrow{\sim} & \mathcal{C}(j) \\ \text{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j} \searrow & & \nearrow \gamma \\ & \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} & \end{array} \quad \begin{array}{ccc} \kappa \otimes \mathcal{C}(j) & \xrightarrow{\sim} & \mathcal{C}(j) \\ \eta \otimes \text{id}_{\mathcal{C}(j)} \searrow & & \nearrow \gamma \\ & \mathcal{C}(1) \otimes \mathcal{C}(j) & \end{array}$$

- *Associativity:* Fix natural numbers $k, j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + \dots + j_s$ for $1 \leq s \leq k$. For each $i_1, \dots, i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)+1}} + \dots + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{array}{ccc} [\mathcal{C}(k) \otimes (\bigotimes_{s=1}^k \mathcal{C}(j_s))] \otimes (\bigotimes_{r=1}^j \mathcal{C}(i_r)) & \xrightarrow{\gamma \otimes \text{id}} & \mathcal{C}(j) \otimes (\bigotimes_{r=1}^j \mathcal{C}(i_r)) \\ \downarrow \text{shuffle} & & \searrow \gamma \\ \mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)+q}}) \right) \right) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_s \gamma)} & \mathcal{C}(k) \otimes (\bigotimes_{s=1}^k \mathcal{C}(h_s)) \\ & & \nearrow \gamma \\ & & \mathcal{C}(i) \end{array}$$

- *Equivariance:*

- Given $\sigma \in \Sigma_k$, $j_1 + \dots + j_k = \Sigma_j$, let $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes “permutation of k -blocks of letters”.

Example 2.3. Given $j = 5 = 2 + 3$, the permutation $(1\ 2) \in \Sigma_2$, $\sigma(2, 3)$ is the permutation $\langle 1, 2; 3, 4, 5 \rangle \mapsto \langle 3, 4, 5; 1, 2 \rangle$ (namely changing the blocks’ order, without changing the order in each block).

Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes (\bigotimes_{r=1}^k \mathcal{C}(j_r)) & \xrightarrow{\sigma \otimes \text{shuffle}_{\sigma^{-1}}} & \mathcal{C}(k) \otimes (\bigotimes_{r=1}^k \mathcal{C}(j_{\sigma(r)})) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})} & \mathcal{C}(j) \end{array}$$

- Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus \dots \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_r th block in j (for instance, let τ_1 acts on $1, \dots, j_1$; let τ_2 acts on $j_1 + 1, \dots, j_1 + j_2$, etc.). Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_r \tau_r)} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) \\ \downarrow \gamma & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{C}(j) \end{array}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \rightarrow \mathcal{C}$ being $\text{id}_{\mathcal{C}(0)}$), the operad is called *unital operad*.

Definition 2.4. Let $\mathcal{C}, \mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi : \mathcal{C} \rightarrow \mathcal{C}'$, is a sequence of Σ_j -equivariant morphisms $\psi_j : \mathcal{C}(j) \rightarrow \mathcal{C}'(j)$ in $\mathcal{V}$ (i.e., $\sigma^{-1} \circ \psi_j \circ \sigma = \psi_j$, for all $\sigma \in \Sigma_j$), such that it preserves the operad's structure morphism:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) & \xrightarrow{\gamma} & \mathcal{C}(j_1 + \dots + j_k) \\ \downarrow \psi_k \otimes \psi_{j_1} \otimes \dots \otimes \psi_{j_k} & & \downarrow \psi_{j_1 + \dots + j_k} \\ \mathcal{C}'(k) \otimes \mathcal{C}'(j_1) \otimes \dots \otimes \mathcal{C}'(j_k) & \xrightarrow{\gamma'} & \mathcal{C}'(j_1 + \dots + j_k) \end{array}$$

2.1 Categorical Example: Associative Operad

Let $\mathcal{M}(j) := \kappa[\Sigma_j] := \coprod_{\psi \in \Sigma_j} \kappa_{\psi}$, coproduct of $\#\Sigma_j$ copies of κ (each κ_{ψ} is the unit κ , subscript is for labeling convenience). It has a natural right Σ_j -action, where for each $\sigma \in \Sigma_j$, the identity $\kappa_{\psi\sigma^{-1}} \xrightarrow{\sim} \kappa_{\psi}$ induces an isomorphism on coproduct $\tilde{\sigma} \in \text{Aut}(\mathcal{M}(j))$:

$$\begin{array}{ccc} \kappa_{\psi} & \xrightarrow{\sim} & \kappa_{\psi\sigma} \\ \downarrow \iota_{\psi} & & \downarrow \iota_{\psi\sigma} \\ \coprod_{\psi \in \Sigma_j} \kappa_{\psi} & \xrightarrow{\exists! \tilde{\sigma}} & \coprod_{\psi \in \Sigma_j} \kappa_{\psi\sigma} \end{array}$$

This construction enforces $(\sigma\tilde{\sigma}') = \tilde{\sigma}' \circ \tilde{\sigma}$, which is a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{M}(j))$.

There is a natural unit morphism $\eta : \kappa \rightarrow \kappa[\Sigma_1] = \kappa$ via identity.

Also, each identity morphism $\kappa_{\psi} \rightarrow \kappa$ induces a trivialization $t : \kappa[\Sigma_j] \rightarrow \kappa$. Combining the action $\sigma \in \Sigma_k$ (by identifying concatenation of blocks $(j_{\sigma^{-1}(1)}, \dots, j_{\sigma^{-1}(k)}) = \{1, \dots, j\}$ in order, also inducing inclusion $\Sigma_{j_s} \hookrightarrow \Sigma_j$), we get the following morphism:

$$\kappa_{\psi} \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \xrightarrow{\sim} \bigotimes_{s=1}^k \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \xrightarrow{t^{k-1}} \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \rightarrow \kappa[\Sigma_j]$$

Running through all $\psi \in \Sigma_k$, this induces a morphism $\gamma : \coprod_{\psi \in \Sigma_k} \left[\kappa_{\psi} \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \right] \rightarrow \kappa[\Sigma_j]$, and use the fact $\otimes$ commutes with colimits, this is equivalent to a morphism $\gamma : \kappa[\Sigma_k] \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \rightarrow \kappa[\Sigma_j]$.

Verify $\mathcal{M}$ forms an operad is a long tedious work, unless many people request the proof we'll omit it here :)

2.2 Topological Example: Stasheff Associahedra

3 Operad Algebra